WORKED SOLUTION

STEP II 2009, Question 8

Pure • Vector Geometry

STEP II 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q8

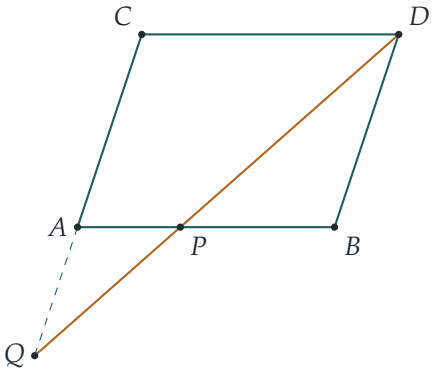
The key idea

Rewrite each definition as "start at one point and move a multiple of a side". Then $\mathbf{p} - \mathbf{b} = \lambda(\mathbf{a} - \mathbf{b})$ and $\mathbf{q} - \mathbf{c} = \mu(\mathbf{a} - \mathbf{c})$, so the lengths BP and CQ are just $\lambda\, AB$ and $\mu\, AC$, and the length condition turns into a very simple equation. To show three points are collinear, show that two of the vectors joining them are parallel.

Solution

The diagram

Since $\mathbf{p} = \mathbf{b} + \lambda(\mathbf{a} - \mathbf{b})$ with $0 < \lambda < 1$, the point P lies on the segment AB, strictly between A and B. Since $\mathbf{q} = \mathbf{c} + \mu(\mathbf{a} - \mathbf{c})$ with $\mu > 1$, the point Q lies on the line CA, beyond A: start at C, go to A, and keep going. The figure (drawn with $\lambda = 0.6$) also shows the point D that appears below.



Finding μ

From the rearrangements above,

$$BP = |\mathbf{p} - \mathbf{b}| = \lambda|\mathbf{a} - \mathbf{b}| = \lambda\, AB, \qquad CQ = |\mathbf{q} - \mathbf{c}| = \mu|\mathbf{a} - \mathbf{c}| = \mu\, AC,$$

using $\lambda > 0$ and $\mu > 0$ to take them out of the modulus. The condition $CQ \times BP = AB \times AC$ becomes $\lambda\mu\, AB \cdot AC = AB \cdot AC$. Since A, B, C are distinct, $AB \cdot AC \neq 0$, so

$$\mu = \frac{1}{\lambda}.$$

This does satisfy $\mu > 1$, because $0 < \lambda < 1$.

The fixed point D

Put $\mathbf{d} = -\mathbf{a} + \mathbf{b} + \mathbf{c}$. We compare $\overrightarrow{PQ}$ and $\overrightarrow{PD}$. First,

$$\begin{aligned}\mathbf{d} - \mathbf{p} &= -\mathbf{a} + \mathbf{b} + \mathbf{c} - \lambda\mathbf{a} - (1-\lambda)\mathbf{b} \\ &= -(1+\lambda)\mathbf{a} + \lambda\mathbf{b} + \mathbf{c}.\end{aligned}$$

Next, with $\mu = 1/\lambda$,

$$\begin{aligned}\mathbf{q} - \mathbf{p} &= \Big(\frac{1}{\lambda} - \lambda\Big)\mathbf{a} + \Big(1 - \frac{1}{\lambda}\Big)\mathbf{c} - (1-\lambda)\mathbf{b} \\ &= \frac{1-\lambda}{\lambda}\Big[(1+\lambda)\mathbf{a} - \lambda\mathbf{b} - \mathbf{c}\Big],\end{aligned}$$

using $\frac{1}{\lambda} - \lambda = \frac{(1-\lambda)(1+\lambda)}{\lambda}$ and $1 - \frac{1}{\lambda} = -\frac{1-\lambda}{\lambda}$.

> **Ask yourself** How do we see that these two vectors are parallel?

The bracket is exactly $-(\mathbf{d} - \mathbf{p})$. So

$$\mathbf{q} - \mathbf{p} = -\frac{1-\lambda}{\lambda}\,(\mathbf{d} - \mathbf{p}), \qquad \text{that is} \qquad \mathbf{d} = \mathbf{p} - \frac{\lambda}{1-\lambda}\,(\mathbf{q} - \mathbf{p}).$$

The coefficient is finite because $\lambda \neq 1$, and $\mathbf{q} \neq \mathbf{p}$ because Q is not on the line AB. Hence D lies on the line PQ. The vector $\mathbf{d}$ involves only $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, so this is the same point for every λ, as required. Because the coefficient $-\lambda/(1-\lambda)$ is negative, P lies between Q and D, as in the figure.

The quadrilateral $ABDC$

We have

$$\mathbf{d} - \mathbf{c} = \mathbf{b} - \mathbf{a} \quad \text{and} \quad \mathbf{d} - \mathbf{b} = \mathbf{c} - \mathbf{a},$$

so $\overrightarrow{CD} = \overrightarrow{AB}$ and $\overrightarrow{BD} = \overrightarrow{AC}$. Opposite sides are equal and parallel, so $ABDC$ is a **parallelogram**, and D is the vertex opposite A. (It is non-degenerate because A, B, C are not collinear.)

Checks

- A geometric cross-check: with $BD \parallel AC$, triangles PAQ and PBD are similar, so $AQ/BD = AP/BP$. Here $AQ = (\mu - 1)AC$, $BD = AC$, $AP = (1 - \lambda)AB$ and $BP = \lambda AB$, giving $\mu - 1 = (1 - \lambda)/\lambda$, which is $\mu = 1/\lambda$ again.
- With $A = (0, 0)$, $B = (4, 0)$, $C = (1, 3)$ and $\lambda = 0.6$ (the figure) we get $P = (1.6, 0)$, $Q = (-\frac{2}{3}, -2)$ and $D = (5, 3)$. The gradients of QP and PD are both $\frac{15}{17}$.
- We also tested 20,000 random triangles in three dimensions. We solved the length condition for μ numerically, and in every case it agreed with $1/\lambda$ to 10^{-13}, with P, Q, D collinear.

Common slips

- Writing $BP = (1 - \lambda)AB$. The coefficient of $\mathbf{b}$ in $\mathbf{p}$ is $1 - \lambda$, but it is the weight at A that measures the distance from B.
- Calling $ABDC$ "a parallelogram" without checking the order of the vertices. Here it is $A \to B \to D \to C$, so AD and BC are the diagonals.

Beyond STEP

Expressions like $\lambda\mathbf{a} + (1 - \lambda)\mathbf{b}$, whose coefficients add up to 1, are *affine combinations*. They are the natural language of points (as opposed to vectors), and collinearity is an affine property: it does not depend on the choice of origin. This leads on to barycentric coordinates and to classical results such as Menelaus' theorem, which describes exactly when a line cuts the three sides of a triangle. Strang's *Introduction to Linear Algebra* develops the vector side, and Siklos's *Advanced Problems in Mathematics* has more STEP-style vector geometry.

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